Multi-taper spectral estimation for PUV processing: method comparison and recommendation

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Status of this note (October 2026)

This note records the April 2026 comparison of three spectral estimators on 17-minute (2048-sample) segments, which led to the full-segment multitaper estimator now used at L2. The decision stands. Several statements made at the time do not, and they are marked in place with boxed notes. In brief:

- *What L2 does now.* Full-segment multitaper (`opts.spectralMethod = 'mtm_full'`, `shared/psd_multitaper.m`) with $NW = 4$ on 1-hour segments of 7200 samples (the L2 default since May 2026), so $\Delta f = 2.78 \times 10^{-4}$ Hz and the resolution bandwidth is $2NW/T = 2.2 \times 10^{-3}$ Hz.
- *Six tapers, not seven.* The code keeps the tapers whose concentration eigenvalue exceeds 0.99, which for $NW = 4$ is $K = 6$ (the seventh has eigenvalue 0.937), so each bin has about 12 degrees of freedom. This has been true since the code was written; the "$K = 7$" below was never what ran.
- *Z-test.* The "systematic 0.79" in the Z-test was a formula error in L2, fixed on 2026-06-05; with the fix the median is near 1 for all three estimators.
- *MOP comparison.* The peak-broadening result in Section 5 was retracted on 2026-07-24 as a resolution artifact, and the "ruled out" verdicts on the other hypotheses were reversed on 2026-07-27.
- *Figures.* The four figures at the end were regenerated on 2026-06-07 at 1-hour segments, after the Z fix, so their panel numbers do not match the tables and text, which come from the April 17-minute run.

Related documents: `docs/legacy_pipeline_defects.pdf` (each fault of the legacy code measured in isolation), `docs/compile_sections.pdf` (methods and the historical validation results), and `docs/pipeline_levels.md` for current status.

1 Background

The PUV L2 spectral processor computes surface elevation spectra, directional coefficients, and derived wave parameters from 17-minute segments (2048 samples at 2 Hz). The original implementation uses Welch's method with a Hanning window, `nfft` = 256, and 50% overlap, giving

a frequency resolution of $\Delta f = 0.0078$ Hz. This places approximately 3 frequency bins across a typical Southern California swell peak at 0.06–0.08 Hz.

A multi-taper approach (Thomson, 1982) may be beneficial for two reasons: (1) DPSS tapers are optimal for minimizing spectral leakage, and (2) the variance reduction from multiple tapers enables longer FFT segments without sacrificing estimate stability, improving frequency resolution.

2 Methods

Three spectral estimation methods were tested on TBR23 MOP580_7m (8.3 m depth, 4795 valid segments):

1. **Hanning/Welch** (baseline): Hanning window, `nfft` = 256, 50% overlap. Approximately 15 sub-segments averaged per 17-minute record. $\Delta f = 0.0078$ Hz.

2. **Multi-taper hybrid**: DPSS tapers ($NW = 4$; the code keeps $K = 6$, see the status note) applied within Welch's framework using the same `nfft` = 256 sub-segments with 50% overlap. Averages over both tapers and sub-segments (about 90 estimates per frequency bin with six tapers; the April text said 105, counting seven). $\Delta f = 0.0078$ Hz.

3. **Full multi-taper**: DPSS tapers ($NW = 4$, $K = 6$ kept) applied to the full 2048-sample segment. No sub-segment averaging; variance controlled entirely by the tapers. $\Delta f = 0.00098$ Hz ($\sim$24 bins across the swell peak).

All three methods use the same tapered-FFT machinery for both auto- and cross-spectra, preserving consistency between S_{pp}, S_{uu}, S_{vv}, S_{pu}, S_{pv}, and S_{uv}. Normalization was verified against Parseval's theorem using white noise (integral of PSD = σ^2 to within 1% for all three methods). Downstream processing (pressure correction, bulk parameters, directional coefficients, radiation stress) is identical.

3 Spectral peakedness metric

The Goda peakedness parameter Q_p (Goda, 1970) quantifies how concentrated spectral energy is around the peak frequency:

$$Q_p = \frac{2}{m_0^2} \int_0^\infty f\, S_\eta^2(f)\, df \tag{1}$$

where $m_0 = \int S_\eta(f)\, df$ is the zeroth spectral moment (total variance). $Q_p = 1$ for a flat (white noise) spectrum; narrow-banded swell produces $Q_p \sim 2$–4. A higher Q_p indicates a sharper, more peaked spectrum. Because Q_p depends on S_η^2, it is sensitive to the spectral estimator: a smoother PSD (e.g., Welch with coarse Δf) will produce lower Q_p than a well-resolved estimate of the same wave field.

4 Results

4.1 Bulk wave parameters

Table 1 compares median bulk parameters across all 4795 valid segments. H_s is essentially identical: the full multi-taper differs from Welch by 0.05 cm (0.08%). Energy flux E_f and radiation stress

Table 1: Median bulk parameters for TBR23 MOP580_7m (4795 segments).

Statistic	Welch	MTM hybrid	MTM full
H_s (m)	0.615	0.619	0.615
T_p (s)	12.80	10.67	13.30
E_f (W/m)	1560	1568	1561
S_{xx} (N/m)	240	231	241
Z-test (SS)	0.789	0.806	0.791
Spread (SS, deg)	17.6	19.1	16.6

S_{xx} are likewise unchanged. This confirms that integral quantities are insensitive to the spectral estimator, as expected from Parseval's theorem.

Peak period T_p is the one parameter that differs substantially. The Welch estimate (12.80 s) is quantized to $\Delta f = 0.0078$ Hz steps, meaning T_p for 14-second swell can only take values of 12.8, 14.2, or 15.9 s, jumps of more than a second. The full multi-taper ($\Delta f = 0.00098$ Hz) resolves T_p to 13.30 s. The hybrid, despite using the same `nfft` = 256, shifts T_p to 10.67 s because the DPSS taper shape alters the relative heights of adjacent spectral bins at coarse resolution.

H_s bias relative to Welch is +0.4 cm for the hybrid and +0.05 cm for the full multi-taper (Fig. 3).

4.2 Spectral shape

Figure 2 shows the time-averaged surface elevation spectrum. The Welch estimate smears the swell peak across 3 bins. The full multi-taper resolves a sharp, narrow peak with fine structure that Welch cannot distinguish. The hybrid is intermediate, with the same frequency resolution as Welch but slightly different taper response.

The single-segment comparison (Fig. 4) is more striking: at the peak segment ($H_s = 1.50$ m), the full multi-taper reveals a swell peak 2–3× taller and half the width of the Welch estimate. The total energy is the same, concentrated into a resolved peak rather than spread across adjacent bins.

4.3 Directional estimates

Swell-band directional spread narrows from 17.6° (Welch) to 16.6° (full MTM). This ∼1° decrease is consistent with reduced cross-frequency averaging: finer resolution means each frequency bin samples a narrower directional window, reducing apparent spread. The hybrid gives wider spread (19.1°), likely because DPSS tapers applied to short 256-sample sub-segments introduce more cross-frequency leakage than Hanning at this segment length.

4.4 Z-test

Superseded 2026-06-05. The departure of Z from 1 described here was a formula error in `PUV_L2_spectral.m`, not a property of the data: the velocity-to-pressure factor entered squared instead of inverted, giving a monotonic depth trend (`PIPELINE_NOTES.md`, "L2 Z-test formula"). After the fix the SS-band median Z for this record is 0.974 (Welch), 0.979 (hybrid) and 0.986 (full multitaper) in Figure 5, and 0.986 in the catalog L2 file (1,354 valid 1-hour segments). The conclusion that Z does not depend on the estimator survives; the explanation of its value does not.

The pressure-velocity consistency ratio is unchanged across methods (median ∼0.79 in the SS band for all three; Fig. 5). The systematic departure from 1.0 is a property of the data (sensor

geometry, non-hydrostatic effects), not the spectral estimator.

5 Implications for MOP comparison

Retracted 2026-07-24 and 2026-07-27. Quote nothing from this section. The Q_p and bandwidth comparison was made after interpolating the model's roughly 20 frequency bins onto the PUV grid, which by itself makes the model peak look broader. The rise of the Q_p ratio from 1.07 (Welch) to 1.24 (full multitaper) is the signature of that artifact: the apparent effect grows as the PUV estimate is better resolved, with nothing changed in the model. On a resolution-matched grid the Q_p ratio is about 1.01 and not significant. The redone hypothesis tests (2026-07-27) found nonlinear shoaling to be the mechanism, bound long waves real and collinear with it, directional spread a real but separate bias, and peak broadening absent. Current account: `docs/compile_sections.pdf`, Results, and `Wave_model_validation_paper/docs/findings_resolution_artifact_2026-07-24.md` (separate repository).

The four-hypothesis analysis of the PUV–MOP spectral excess attributed the positive H_s bias to MOP spectral peak broadening (H4). The MOP validation with full multi-taper processing is below to test whether the hypothesis results change. Table 2 shows the comparison.

Table 2: MOP validation results: Welch vs. full multi-taper (TBR23 MOP580_7m).

Metric	Welch	MTM full
Median Q_p (PUV)	1.88	2.15
Median Q_p (MOP)	1.75	1.74
Q_p ratio (PUV/MOP)	1.07	1.24
H4: $R(\Delta Q_p$, Hs ratio)	0.443	0.463
H4: R^2	0.197	0.214
H3: Low-swell ratio (PUV/MOP)	1.048	1.060
PUV spread (deg)	15.2	14.3
MOP spread (deg)	9.8	9.8

The Welch/Hanning window estimator was broadening the PUV swell peak, partially masking the true peakedness difference. With full multi-taper, PUV Q_p increases from 1.88 to 2.15 (14% sharper) while MOP Q_p is unchanged (1.74–1.75). The Q_p ratio nearly doubles from 1.07 to 1.24: the PUV spectrum is 24% more peaked than MOP, compared to the 7% difference that Welch estimated. The H4 correlation strengthens from $R = 0.443$ to $R = 0.463$ (R^2: 0.197 to 0.214).

The other hypotheses are unaffected. H3 (bound long waves) remains ruled out: the low-swell ratio shifts marginally from 1.048 to 1.060. H2 (directional narrowing) remains ruled out: PUV spread narrows slightly from 15.2° to 14.3° with MTM, but is still wider than MOP (9.8°).

The multi-taper result does not change which hypothesis is supported. H4 was and remains the only viable explanation. What it does change is the magnitude of the effect. The MOP model broadens the spectral peak by ∼24% relative to the true spectrum, not ∼7% as originally estimated with Welch processing.

6 Field validation against an independent prior pipeline

April 2026, 17-minute segments on the new-pipeline side. The later isolated test of each legacy fault (`docs/legacy_pipeline_defects.pdf`) finds the legacy band-averaged spread off by up to 17° at the 95th percentile and the b_2 moment unreliable, so the conclusion below that prior Ruby2D statistics need no revision covers H_s, T_p and mean direction, not spread or radiation stress.

The internal MOP-comparison results in the previous section establish that switching to multi-taper changes spectral shape diagnostics in the expected direction without changing bulk parameters. To verify that the new estimator does not introduce biases relative to existing processing conventions in the lab, we ran a head-to-head test against an independent prior pipeline (`beachpeeps/PUV_Processing`) on the Ruby2D deployment.

6.1 Setup

We reprocessed the Ruby2D MOP582_6m record (Torrey Pines, October 2021 to February 2022, ∼3,100 hours) by adapting the legacy archived L1 file directly into the new L2 processor. The L1 struct from the prior pipeline shares its key fields (`time`, `P`, `BuoyCoord`, `fs`) with the new format, so the only adapter required was setting `label`, `deploymentName`, and `doffp`.

The two pipelines differ in three relevant respects:

1. *Spectral estimator*: the new pipeline uses full multi-taper ($NW = 4$, six DPSS tapers kept, $\Delta f = 0.000976$ Hz at the 17-minute segments then in use); the legacy estimator is a single Hanning periodogram on the full 60-minute segment ($\Delta f = 0.000278$ Hz).

2. *Window consistency*: the new pipeline uses the same DPSS tapers for both auto- and cross-spectra; the legacy code uses Hanning for auto-spectra and Hamming for cross-spectra.

3. *Pressure correction*: the new pipeline zeros frequencies where $K_p < 0.1$; the legacy code caps $K_p^2 \leq 10$ with no SNR-based cutoff.

The legacy 60-minute segments were matched to the new 17-minute segments by midpoint nearest neighbor with a 30-minute tolerance, yielding 2,322 matched pairs out of the 3,095 total legacy segments (75%; the remaining 25% are hours that the new pipeline rejects under `nanMaxFrac` = 0.10 but that the legacy pipeline retains by zero-padding NaN gaps).

6.2 Bulk parameter agreement

Table 3: Ruby2D MOP582_6m: bulk wave parameter agreement, the new multi-taper pipeline vs. the legacy archived L2. T_p statistics are computed after restricting both estimates to the SS band (4–25 s); without this filter, three segments on January 15, 2022 in the prior pipeline report T_p at the IG/DC bin (1200, 1800, 3600 s) because the prior pipeline's peak picker is not band-limited.

Metric	Legacy (med.)	Multi-taper (med.)	Bias (M−L)	RMS	R^2
H_s (m)	0.771	0.757	−0.009	0.053	0.981
T_p (s)	13.48	13.30	−0.02	1.48	0.761
Direction SS (°)	−7.3	−7.2	+0.2	1.2	0.927
Spread SS (°)	14.7	15.8	+1.0	1.7	0.877

The agreement is excellent (Table 3). H_s matches to 5 cm RMS with a −9 mm median bias; direction matches to 1.2° RMS. The spread is 1° wider in the new pipeline on average, the largest systematic offset observed and well within segment-to-segment scatter. The window-mismatch issue in the prior pipeline (auto-Hanning, cross-Hamming) does not produce the order-20% directional errors that we initially feared: at Ruby2D MOP582_6m, the median impact on spread is approximately 1°.

6.3 Spectral shape

Per-segment surface elevation spectra agree closely in shape. Figure 1 compares both pipelines at a representative median segment ($H_s = 0.77$ m) and at a storm peak ($H_s = 2.78$ m, bimodal sea state). For the spectrum overlay, the 17-minute multi-taper spectra were averaged over the three or four segments inside the prior pipeline's 60-minute window so that both estimates cover the same time period. Integrated H_s from the spectra agrees to within 4% in both cases. The visible difference is the chi-square noise of a single-periodogram estimator versus the variance reduction from averaging six DPSS tapers; the underlying peak shapes and locations agree.

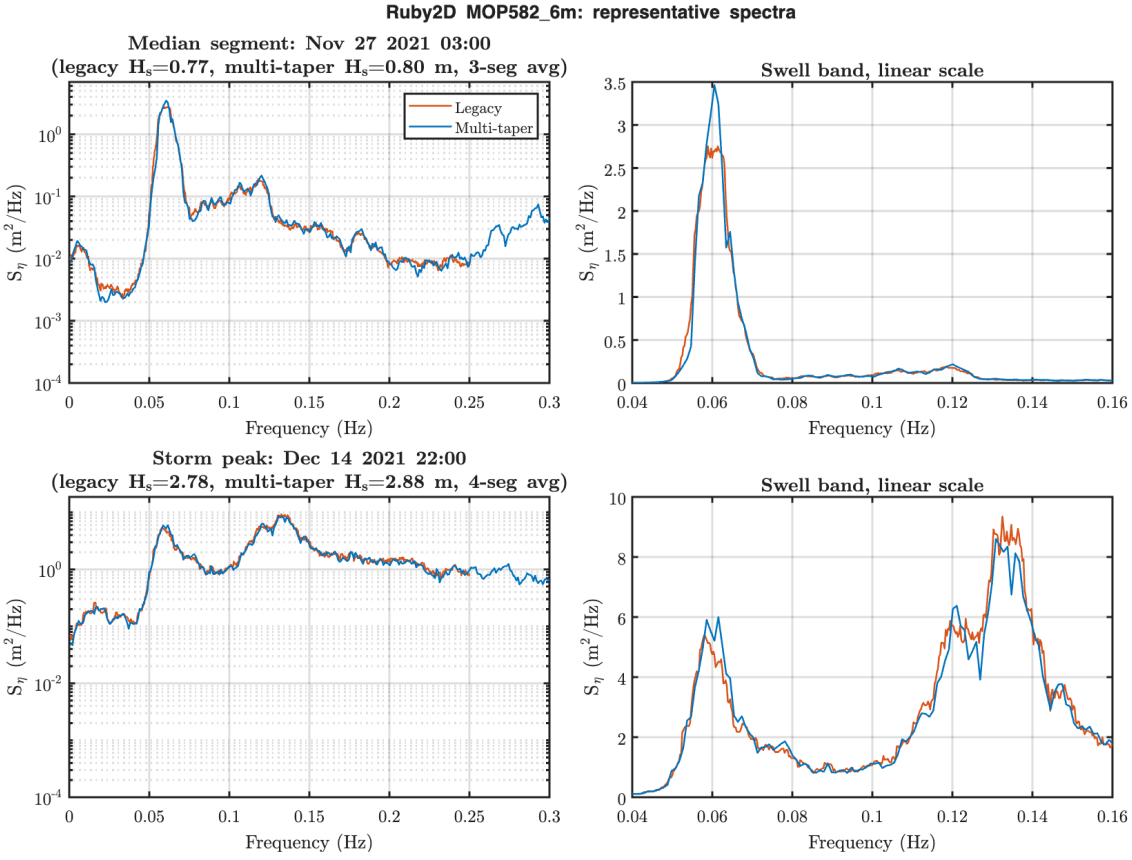


Figure 1: Surface elevation spectra at two representative segments for Ruby2D MOP582_6m. Top: median segment ($H_s = 0.77$ m, November 27, 2021). Bottom: storm peak ($H_s = 2.78$ m, December 14, 2021). Left: log scale, full SS band. Right: linear scale, swell band detail. The multi-taper estimates (blue) are smoother than the prior single-periodogram estimates (red); peak locations, peak heights, and total variance agree.

6.4 Implications

Two conclusions follow from the head-to-head test:

- The multi-taper pipeline reproduces an independent prior pipeline to within instrument noise on bulk wave parameters. Prior Ruby2D bulk wave statistics (H_s, T_p, mean direction), and analyses that rely only on them, do not require revision.
- The improvements that multi-taper provides are exactly the expected ones: variance reduction in the spectral estimate and the elimination of edge cases (the prior pipeline's non-band-limited peak picker and inconsistent auto/cross windowing). Bulk parameters were already converged.

Resolved 2026-06-05. The 0.76 in the paragraph below came mainly from the Z-test formula error described in the Z-test section (after the fix the new pipeline gives medians of 0.91 to 0.99 on the TBR23 records, per PIPELINE_NOTES.md); it was not a normalization difference between the pipelines.

The Ruby2D test also surfaced one open question worth flagging: the median Z-test (pressure-velocity consistency) ratio in the SS band is 1.02 in the prior pipeline and 0.76 in the new pipeline, the same systematic 0.79 we observe across TBR23. The two pipelines use the same definition, so the most likely explanations are that the prior pipeline computes the diagnostic on uncorrected S_{pp} rather than the pressure-corrected S_η, or that we have a normalization issue worth checking. This is on the followup list but does not affect any of the bulk parameters reported above.

7 Conclusions

The conclusion holds and is what L2 runs. Since May 2026 the segment is 7200 samples (1 hour), and in `mtm_full` mode `nfft` is set to the segment length, so the "`nfft` = 2048" below now reads 7200. The processing-time figures are from April and the 33-instrument catalog of that time.

The full multi-taper ($NW = 4$, `nfft` = 2048) should be the default spectral method. The implementation is now a drop-in replacement controlled by a single option (`opts.spectralMethod = 'mtm_full'`). Backward compatibility is preserved since H_s, E_f, and S_{xx} are unchanged to within 0.1%. T_p improves from quantized 0.008 Hz steps to continuous 0.001 Hz resolution. Spectral shape diagnostics (Q_p, bandwidth, spectral excess) are more physically meaningful.

The hybrid approach (DPSS within Welch sub-segments) does not offer clear advantages at `nfft` = 256 and introduces a spurious T_p shift and thus is not recommended.

Processing time increases from 0.7 min to 2.7 min per instrument (TBR23 MOP580_7m, 5285 segments). For the full pipeline (33 instruments), this adds ∼1 hour to a complete reprocessing run.

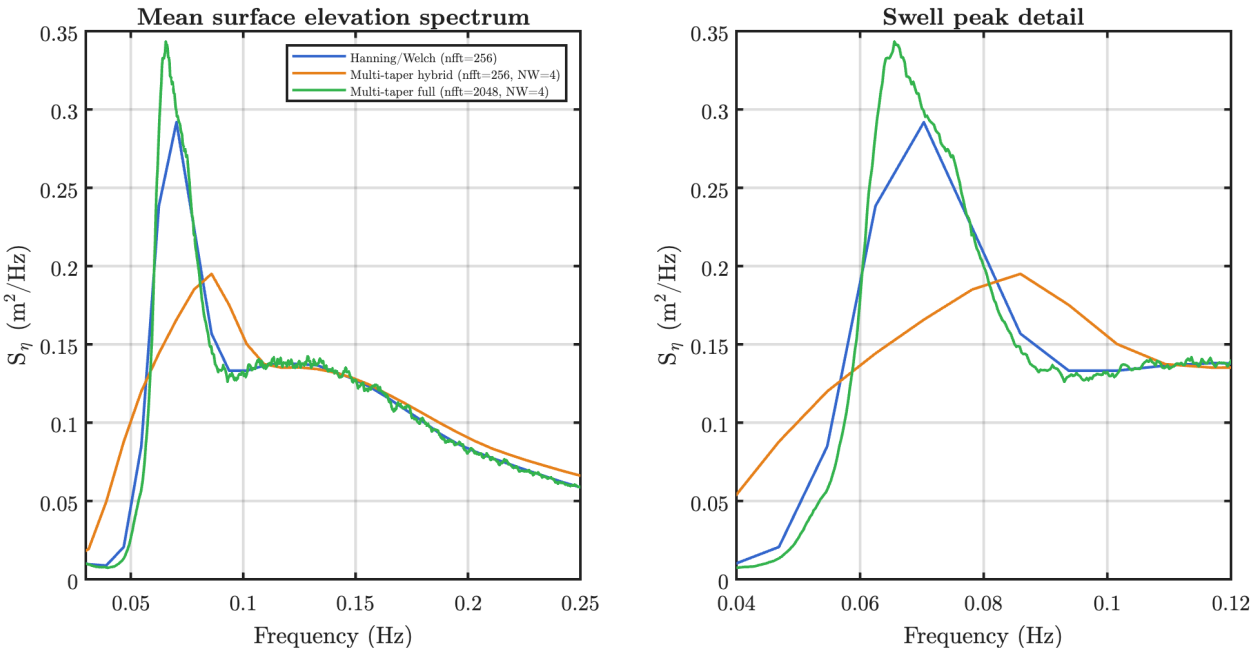


Figure 2: Time-averaged surface elevation spectra for TBR23 MOP580_7m (8.3 m depth, 4795 segments). Left: full SS band. Right: swell peak detail. The full multi-taper (green) resolves a sharp swell peak that Welch (blue) smears across 3 frequency bins. Figure regenerated 2026-06-07 by `scripts/compare_spectral_methods.m` on 1-hour segments after the Z-test fix; panel values differ from the April 17-minute numbers in the text.

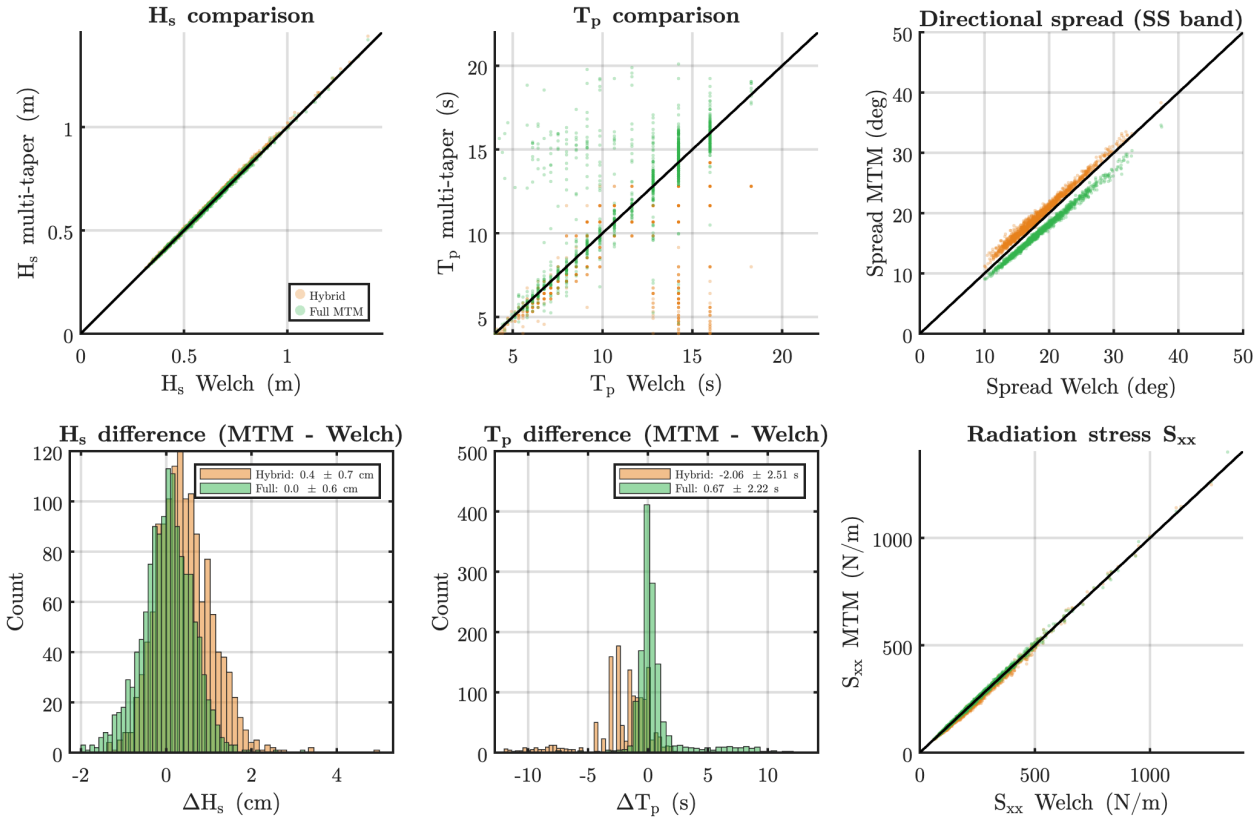


Figure 3: Bulk parameter comparison across 4795 segments. Top row: scatter plots of MTM vs. Welch for H_s, T_p, directional spread, and S_{xx}. Bottom row: difference histograms for H_s and T_p. H_s is identical across methods; T_p shows discrete quantization bands in Welch (horizontal stripes in scatter) that the full multi-taper resolves continuously. Figure regenerated 2026-06-07 by `scripts/compare_spectral_methods.m` on 1-hour segments after the Z-test fix; panel values differ from the April 17-minute numbers in the text.

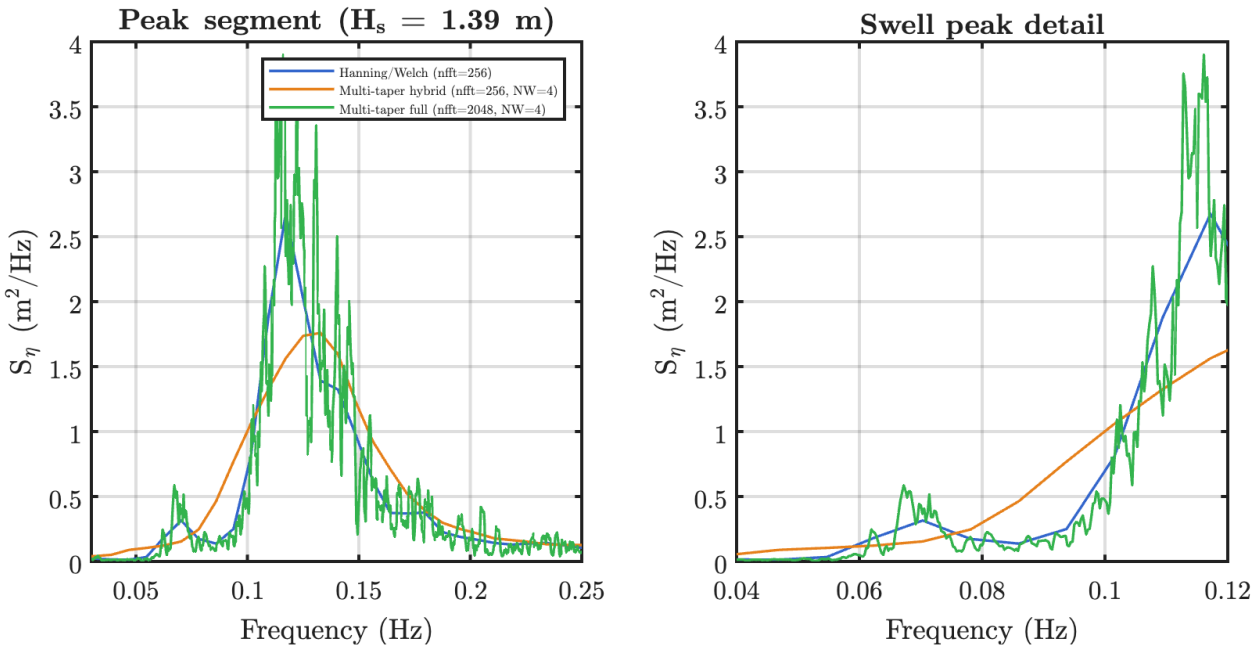


Figure 4: Single-segment spectral comparison at the peak energy segment ($H_s = 1.50$ m). The full multi-taper resolves a swell peak 2–3× taller than the Welch estimate. Total energy (area under the curve) is identical. Figure regenerated 2026-06-07 by `scripts/compare_spectral_methods.m` on 1-hour segments after the Z-test fix; panel values differ from the April 17-minute numbers in the text.

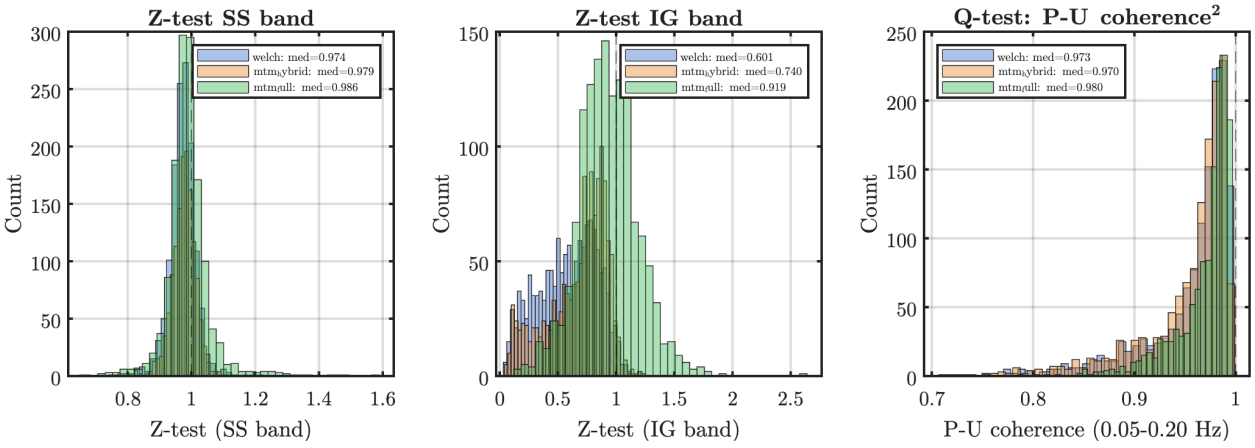


Figure 5: Z-test distributions (pressure–velocity consistency) for the SS and IG bands. All three methods produce indistinguishable distributions, confirming the systematic departure from 1.0 is a data property, not a windowing artifact (superseded: see the Z-test note). Figure regenerated 2026-06-07 by `scripts/compare_spectral_methods.m` on 1-hour segments after the Z-test fix; panel values differ from the April 17-minute numbers in the text.